String Operators and Functions

Lecture Notes · Game DB Junior

In this lesson

- How to join two strings with the || operator
- How to join strings with the **CONCAT()** function
- The functions **UPPER()**, **LOWER()** and **LENGTH()**
- How to **stack** functions inside each other

1. Operators and functions for strings

We can use operators to work with numbers. In the same way, there are operators we can use to work with **strings**. There is also something else we can use: **functions**. Functions in SQL work very much like functions in many other programming languages.

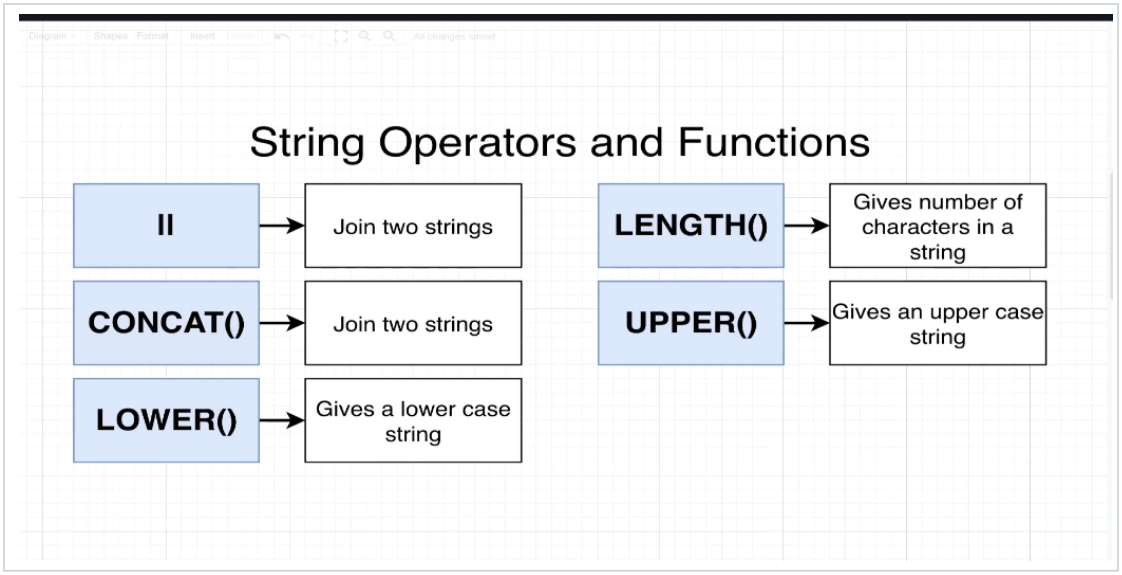


Figure 1 — String operators and functions (a partial list)

Operator / Function	What it does
||	Join two strings
CONCAT()	Join two strings
LOWER()	Gives a lower case string
UPPER()	Gives an upper case string
LENGTH()	Gives the number of characters in a string

2. Joining strings with ||

We want to select the name and country of every city and join (concatenate) them together. We use the **pipe** character two times: ||.

Note: These are not the letter L or the letter I. It is the pipe character. On the keyboard it is just above the return (Enter) key.

```
SELECT name || country FROM cities;
```

The name and country are joined directly together. This result is hard to read.

Adding a comma and a space

We can also join with **string constants**. After the first ||, we put single quotes with a comma and a space inside: **', '**. Then another ||.

```
SELECT name || ', ' || country FROM cities;
```

Read it like this: give me the name, then a comma and a space, then the country. Now we get "Tokyo, Japan". This is easier to read.

Renaming with AS

Just like with math operators, Postgres names this column **?column?**. So we use **AS** to give it a clear name:

```
SELECT name || ', ' || country AS location FROM cities;
```

3. Joining strings with CONCAT()

The function **CONCAT** also joins strings. The || operator and CONCAT do the **exact same thing**. Only the syntax is different.

- We call CONCAT like a function in other programming languages.
- Inside the parentheses we put **arguments**: the strings we want to join, separated by commas.

```
SELECT CONCAT(name, country) FROM cities;
```

This gives the same result as name || country. To add a comma and a space again, we put **', '** as another argument between name and country:

```
SELECT CONCAT(name, ', ', country) AS location FROM cities;
```

The syntax is a bit hard to read, but it means: take the name, a comma and a space, and the country, and join these three strings together.

4. UPPER() and LOWER()

UPPER() gives the upper case value of a column. **LOWER()** gives the lower case value. We can **stack** as many functions together as we want.

Way 1: UPPER on each column, then CONCAT

We wrap name with UPPER( ), and we do the same for country:

```
SELECT
    CONCAT(UPPER(name), ', ', UPPER(country)) AS location
FROM
    cities;
```

The query got long, so the **format** button at the top was used to make it easier to read. When we run it, we get everything in capital letters.

Way 2: UPPER on the result of CONCAT

Instead, we can wrap the whole CONCAT with one UPPER( ) call. This gives the **exact same result**.

```
SELECT
    UPPER(CONCAT(name, ', ', country)) AS location
FROM
    cities;
```

LOWER() works in a very similar way. It gives the lower case version of a string.

5. LENGTH()

LENGTH() tells us the **number of characters** in a string. It may be one of the more useful functions, for example if we want to find strings of a certain length. We will see examples later.

6. Running the code (my practice)

Query 1: join name and country with || and show it as location.

```
-- Combine the city name and country and show it as "location"
SELECT name ||  ', ' || country AS location FROM cities;
```

Figure 2 — Joining with || and renaming with AS

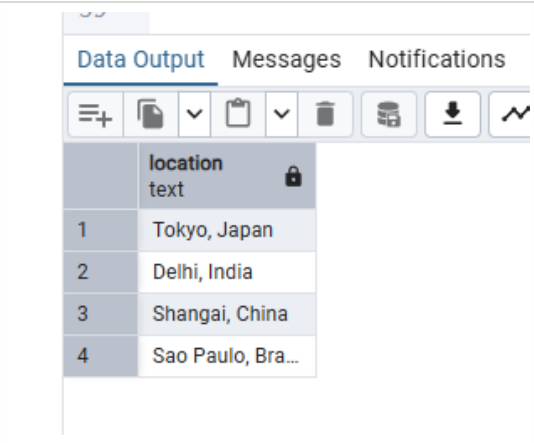


Figure 3 — Result: Tokyo, Japan ...

Query 2: the same thing with the CONCAT function. The result is the same.

```sql
-- Combine the city name and country using the CONCAT function
SELECT CONCAT(name, ', ', country) AS location FROM cities;
```

Figure 4 — Joining with CONCAT()

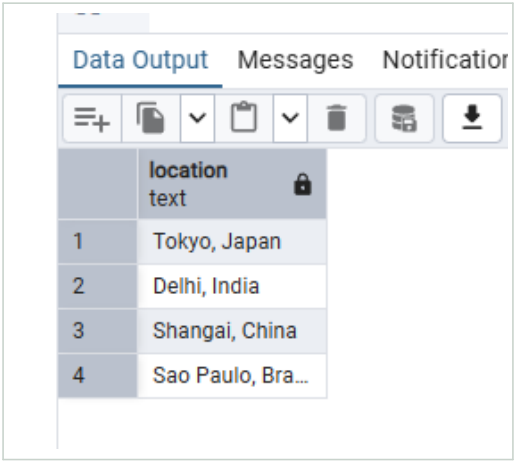


Figure 5 — Result: the same as with ||

Query 3: UPPER on name and on country, then CONCAT.

```sql
-- Combine the city name and country in uppercase and show it as "location"
SELECT
    CONCAT(UPPER(name), ', ', UPPER(country)) AS location
FROM
    cities;
```

Figure 6 — UPPER() inside CONCAT()

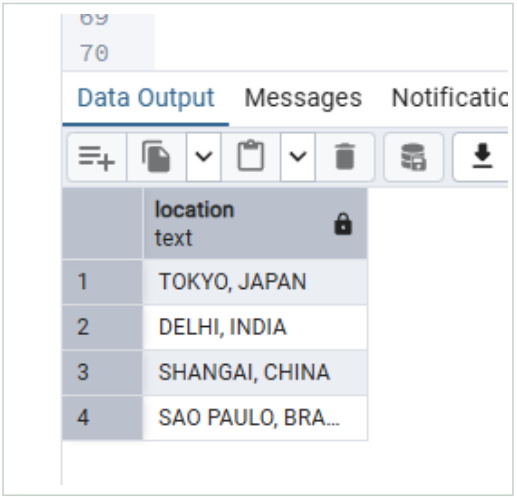


Figure 7 — Result: everything in capital letters

Query 4: UPPER on the whole CONCAT. This gives the same result as Query 3.

```
SELECT
    UPPER(CONCAT(name, ', ', country)) AS location
FROM
    cities;
```

Figure 8 — CONCAT() inside UPPER()

7. Summary

Idea	Remember
a || b	Operator that joins two strings. || is the pipe character two times
CONCAT(a, b, ...)	Function that joins strings. Same result as ||, different syntax
', '	A string constant. We can join it like any other string
UPPER() / LOWER()	Give the upper case / lower case version of a string
LENGTH()	Gives the number of characters in a string
Stacking	We can put functions inside each other, e.g. UPPER(CONCAT(...))
AS	Gives a clear name instead of ?column?

Sources
Lecture transcript and lecture image (Figure 1)
My own practice screenshots (Figures 2–8)